



Certificate of Calibration

Ocean Sonics, Ltd.

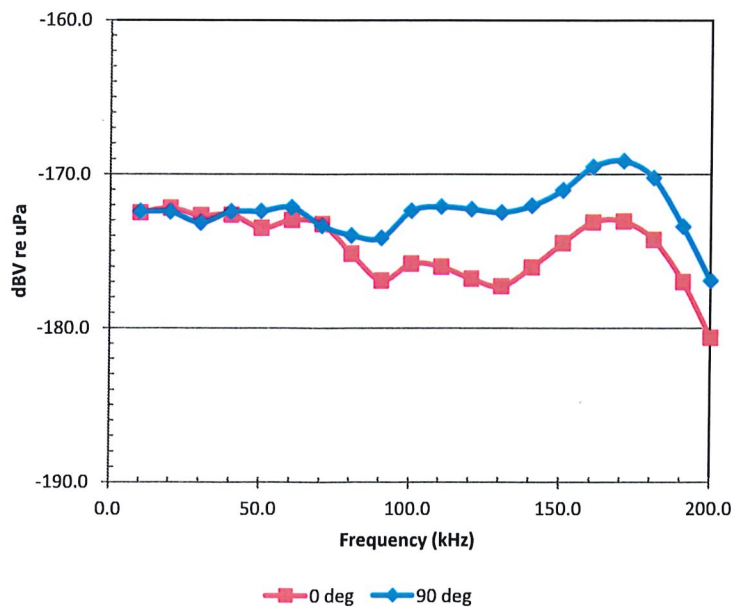
Calibration Certificate Number: **C11040**

Test Result: 10 kHz to 100 kHz: -174.5 ± 2.4
10 kHz to 200 kHz: -174.9 ± 5.7

Model Number SB35-ETH
Serial Number 1248
Manufacture Date 20-Mar-2013
Measurement Date 26-May-2026
Certificate Date 26-May-2026
Sensitivity @ 26 Hz -170.4 dB re V / μ Pa
Case Type Titanium
Element Manufacturer Geospectrum
Element Model M24HF
Element Serial H000957
Preamp Model 04-300435-09
Calibrated By D.Lee
Work Order Number W0-90
Test Type RX Sensitivity
Test Procedure Complex RMS
Test Location Tank #3, 1 m
Water Temperature 16 °C

Projector Manufacturer Ocean Sonics
Projector Model TH2-SER-4F
Projector Serial 2225
Measurement Distance 1 m
Output Level 129.7 dB re μ Pa @ 1 m
Tone Burst 100.0 μ s / 300 ms
Reference Manufacturer Ocean Sonics
Reference Model RB9-ETH
Reference Serial 6465
Primary Calibration 2023-07-04
Preamp Manufacturer Ocean Sonics
Preamp Model 04-300451-01
Preamp Serial Z88
Preamp Gain 36 dB
ADC Manufacturer Ocean Sonics
ADC Model Number 04-300427-01
ADC Serial Number Z87

Receive Sensitivity



Frequency kHz	Sensitivity [dBV re μ Pa]	
	0 deg	90 deg
10.0	-172.5	-172.4
20.1	-172.2	-172.5
30.1	-172.7	-173.2
40.2	-172.6	-172.5
50.2	-173.5	-172.4
60.2	-173.0	-172.2
70.3	-173.2	-173.4
80.3	-175.2	-174.0
90.4	-176.9	-174.2
100.4	-175.8	-172.4
110.4	-176.0	-172.1
120.5	-176.8	-172.3
130.5	-177.3	-172.5
140.5	-176.0	-172.1
150.6	-174.5	-171.1
160.6	-173.1	-169.6
170.7	-173.1	-169.2
180.7	-174.3	-170.3
190.7	-177.0	-173.4
200.0	-180.6	-177.0



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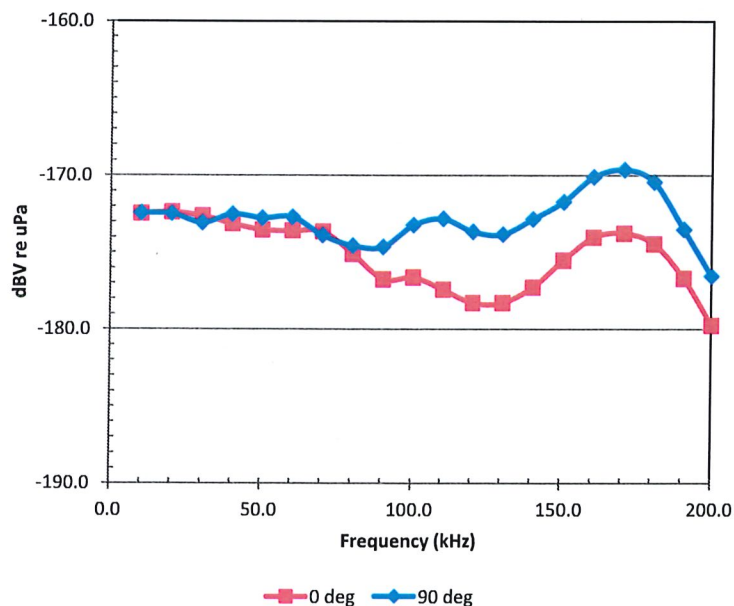
Calibration Certificate Number: **C10908**

Test Result: 10 kHz to 100 kHz: -174.6 ± 2.2
10 kHz to 200 kHz: -174.7 ± 5.0

Model Number SB35-ETH
Serial Number 1248
Manufacture Date 20-Mar-2013
Measurement Date 27-Apr-2026
Certificate Date 27-Apr-2026
Sensitivity @ 26 Hz -170.4 dB re V / μ Pa
Case Type Titanium
Element Manufacturer GeoSpectrum
Element Model M24HF
Element Serial H000957
Preamp Model 04-300435-09
Calibrated By D.Lee
Work Order Number W0-90
Test Type RX Sensitivity
Test Procedure Complex RMS
Test Location Tank #3, 1 m
Water Temperature 16 °C

Projector Manufacturer Ocean Sonics
Projector Model TH2-SER-4F
Projector Serial 2225
Measurement Distance 1 m
Output Level 129.9 dB re μ Pa @ 1 m
Tone Burst 100.0 μ s / 300 ms
Reference Manufacturer Ocean Sonics
Reference Model RB9-ETH
Reference Serial 6465
Primary Calibration 2023-07-04
Preamp Manufacturer Ocean Sonics
Preamp Model 04-300451-01
Preamp Serial Z88
Preamp Gain 36 dB
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30.1	-172.7	-173.1
40.2	-173.2	-172.5
50.2	-173.6	-172.8
60.2	-173.6	-172.7
70.3	-173.7	-173.9
80.3	-175.1	-174.6
90.4	-176.8	-174.7
100.4	-176.6	-173.3
110.4	-177.4	-172.9
120.5	-178.3	-173.7
130.5	-178.3	-173.9
140.5	-177.3	-172.9
150.6	-175.5	-171.8
160.6	-174.0	-170.1
170.7	-173.8	-169.7
180.7	-174.5	-170.5
190.7	-176.7	-173.6
200.0	-179.7	-176.6